

DEPARTMENT OF COMPUTER ENGINEERING

CSD415 PROJECT PHASE I

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Project Approval Form

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1.AI-Powered Multimodal Misinformation Detection and Verification System

The rapid growth of digital media and social networking platforms has significantly increased the spread of digital misinformation, including fake news, manipulated images, and AI-generated (deepfake) content. Such misleading information can influence public opinion, spread false medical advice, promote financial scams, damage reputations, and create panic during emergencies. Existing fact-checking methods are largely manual, time-consuming, and unable to keep pace with the enormous volume of online content generated every day. As AI-generated content becomes increasingly realistic, distinguishing genuine information from manipulated content has become more challenging. This issue affects a wide range of stakeholders, including the general public, journalists, fact-checking organizations, government agencies, educational institutions, social media users, and researchers, highlighting the need for an efficient and automated solution.

To address this problem, the proposed project aims to develop an **AI-Powered Multimodal Misinformation Detection and Verification System** that utilizes Machine Learning, Deep Learning, Natural Language Processing (NLP), and Computer Vision to analyse textual and visual content. The system will detect misleading news articles and manipulated images, generate a credibility score indicating the reliability of the content, and provide explainable predictions to help users understand the decision-making process. Developed using technologies such as Python, FastAPI, React.js, PostgreSQL, PyTorch, Transformer-based models (BERT/DistilBERT), and OpenCV, the system will assist users in identifying misinformation quickly and accurately. It is expected to support journalists and fact-checkers, improve public awareness and digital literacy, and provide a scalable framework that can be extended to detect deepfake videos and support multilingual misinformation detection in the future.

Reference: <https://www.sciencedirect.com/science/article/abs/pii/S1568494620309881>
<https://link.springer.com/article/10.1007/s13042-025-02633-w>

2. Malware Classification Using Machine Learning with Explainable AI and Visualization Dashboard

Malware is one of the most significant cybersecurity threats, capable of damaging computer systems, stealing sensitive information, disrupting services, and causing substantial financial losses to individuals and organizations. Traditional antivirus solutions primarily rely on signature-based detection techniques, which are effective only against known malware and often fail to identify newly developed or modified variants. With the rapid evolution of cyberattacks, there is a growing need for intelligent malware detection methods that can accurately classify unknown malware based on its characteristics rather than predefined signatures. Furthermore, although Machine Learning models have demonstrated high accuracy in malware classification, many of them function as "black boxes," providing predictions without explaining the reasoning behind them. This lack of transparency reduces user confidence and makes it difficult for cybersecurity professionals to validate and trust the model's decisions.

To address these challenges, the proposed project aims to develop a **Machine Learning-Based Malware Classification System with Explainable AI and an Interactive Visualization Dashboard**. The system will accept malware samples or extracted features as input, classify them into their respective malware families using a trained Machine Learning model, and present the prediction along with a confidence score. Explainable AI techniques, such as SHAP, will be integrated to provide meaningful explanations for the model's predictions, helping users understand which features influenced the classification. An interactive dashboard developed using React.js and visualization libraries such as Chart.js or Plotly will display malware analysis results, trends, and performance metrics in a user-friendly manner. Implemented using technologies including Python, Scikit-learn, Flask, React.js, and Explainable AI frameworks, the system is expected to improve malware detection accuracy, enhance transparency and trust in Machine Learning models, support faster decision-making for cybersecurity professionals, and serve as an educational platform for understanding the practical application of Artificial Intelligence in cybersecurity.

Reference: <https://ieeexplore.ieee.org/document/10968061>

3.AI-Powered Cloud Security Operations Center (SOC) Using Spring Boot and Agentic AI for Intelligent Threat Detection and Security Alert Management

With the widespread adoption of cloud computing, web applications, and enterprise software, organizations generate massive volumes of security logs from user authentication, application activities, network traffic, and database operations. Security analysts working in a Security Operations Center (SOC) are responsible for continuously monitoring these logs to identify cyber threats such as brute-force attacks, SQL injection attempts, malware activity, and unauthorized access. However, traditional SOC's rely heavily on manual analysis of thousands of security alerts, making the process time-consuming, repetitive, and prone to human error. As the number of security events continues to grow, genuine threats may be overlooked while analysts spend valuable time investigating false positives, resulting in delayed incident response, increased operational costs, and greater cybersecurity risks. This challenge affects organizations, cloud service providers, IT administrators, cybersecurity professionals, educational institutions, and researchers who require efficient and intelligent security monitoring solutions.

To address these challenges, the proposed project aims to develop an **AI-Powered Cloud Security Operations Center (SOC) Using Spring Boot and Agentic AI for Intelligent Threat Detection and Security Alert Management**. The system will securely collect, store, and analyse security logs through an interactive web-based dashboard while utilizing an Agentic AI module to automatically detect suspicious activities, classify threats into different severity levels, generate AI-based explanations, and recommend appropriate response actions. Built using technologies such as Java, Spring Boot, Spring Security, MySQL, REST APIs, HTML, CSS, JavaScript, Bootstrap, Docker, Kubernetes, and Git, the application will provide real-time monitoring, intelligent alert management, and automated security reporting. The proposed solution is expected to reduce the workload of security analysts, improve the speed and accuracy of threat detection, strengthen organizational cybersecurity, and demonstrate the practical application of Artificial Intelligence and cloud technologies in modern Security Operations Centers while also serving as a valuable educational platform for students and researchers in cybersecurity and software engineering.

Reference: <https://ieeexplore.ieee.org/document/10850912>